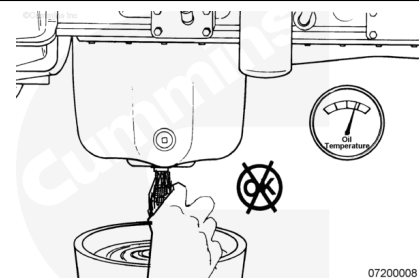


Trainings ▾

Drain

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

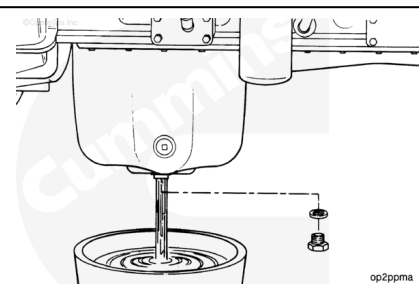
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Change the lubricating oil and filter at the specified oil change interval. Refer to Procedure 102-002 in Section 2 (</qs3/pubsys2/xml/en/procedures/35/35-102-002-om-ind.html>) to find the correct change interval for your application.

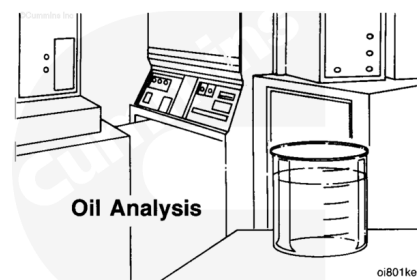
Operate the engine until the water temperature reaches 60°C [140°F]. Shut off the engine. Remove the oil drain plug from the bottom of the lubricating oil pan. Do **not** remove the plugs on either side of the oil pan to drain the oil. They will **not** allow the oil to drain completely.

Operate the engine until the water temperature reaches 60°C [140°F]. Shut the engine OFF. Remove the oil drain plug from the bottom of the lubricating oil pan.

Note : Fittings used in the bottom drain opening of the oil pan other than Cummins-specified parts must not exceed the size or weight limits. Refer to Procedure 018-017 in Section V. (</qs3/pubsys2/xml/en/procedures/35/35-018-017-om-ind.html>)



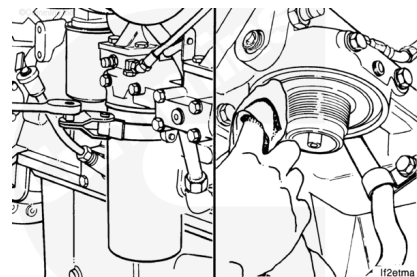
On standby generator applications, Cummins Inc. recommends oil sampling and analysis at the time of oil change to monitor oil contaminant levels.



Clean the area around the lubricating oil filter head.

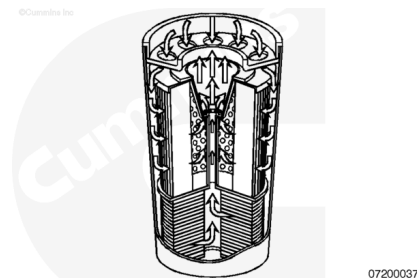
Use oil filter wrench, Part Number 3375049, to remove the oil filter(s).

Clean the gasket surface of the oil filter head. The o-ring can stick on the filter head; **make sure** it is removed.



Fill

Use the correct oil filter for your engine. Refer to Procedure 102-002 in Section 2.
(</qs3/pubsys2/xml/en/procedures/35/35-102-002-om-ind.html>)

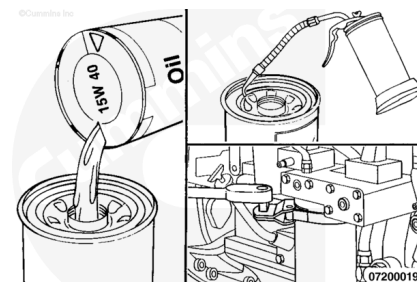


⚠ CAUTION ⚠

The lack of lubrication, caused by an empty oil filter, is harmful to the engine.

⚠ CAUTION ⚠

Mechanical overtightening of the filter can distort the threads or damage the filter element seal.



Fill the filter with clean oil. Choose the correct oil for your operating climate as outlined in Section V. Apply a light film of clean engine oil to the gasket sealing surface before installing the new filter.

Install the filter on the filter head. Tighten the filter until the gasket contacts the filter head surface.

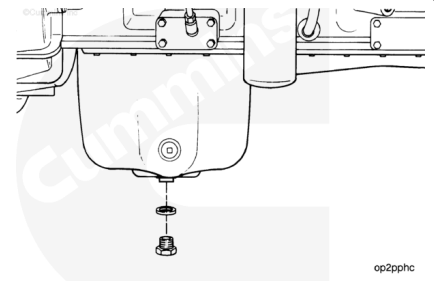
Use oil filter wrench, Part Number 3375049, to tighten the filter to the specifications supplied with the filter.

Clean and check the oil drain plug threads and the seal surface.

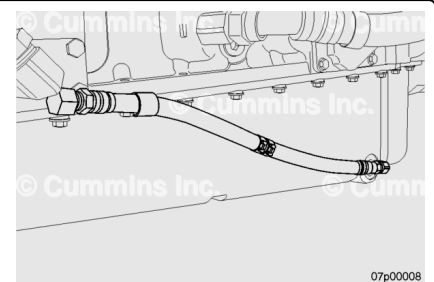
Install the oil drain plug in the lubricating oil pan.

Torque Value: 88 n•m [65 ft-lb]

Note : Check the torque of the side and bottom drain plugs to be sure both are tight.



If a remote drain hose has been provided, and a change oil pump used, disconnect the hose from the pump and install the cap onto the end of the hose and secure the hose to the side of the lubricating oil pan using the clips that are provided.

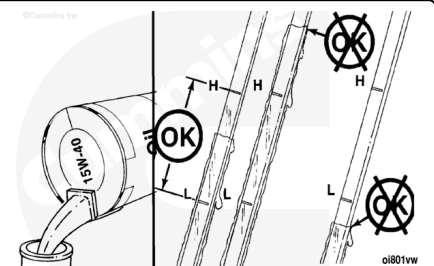


Use a high-quality multiviscosity oil such as Valvoline® Premium Blue™, Valvoline® Premium Blue™ 2000, or equivalent. Choose the correct oil for a specific operating climate. Refer to Procedure 018-003 in Section V.
(/qs3/pubsys2/xml/en/procedures/101/101-018-003-om-ind.html)



Fill the engine with clean oil to the H (high) mark on the dipstick. For total system capacity including filter, Refer to Procedure 018-017 in Section V.
(/qs3/pubsys2/xml/en/procedures/35/35-018-017-om-ind.html)

Note : Check the Centinel™ makeup tank oil level, if equipped.

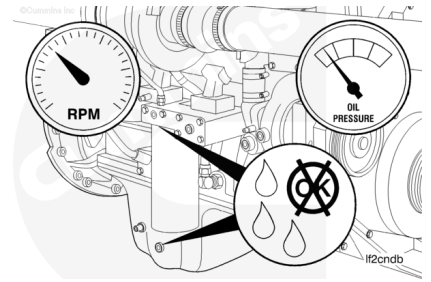


⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine

damage. Confirm the correct oil level is in the oil pan.

Operate the engine at idle speed to inspect for leaks at the filter and the drain plugs.



Shut off the engine. Wait approximately 10 minutes for the oil to drain back from the upper parts of the engine to the oil pan.

Check the oil level again. Add oil as necessary to bring the level up to the "H" (high) mark on the dipstick.

